JARAMOGI OGINGA ODINGA UNIVERSITY OF SCIENCE AND TECHNOLOGY

OFFICE OF THE SENIOR LECTURER / LEAD RESEARCHER

REQUEST FOR APPROVAL

TO:	**The Vice-Chancellor**, Jaramogi Oginga Odinga University of Science and Technology (JOOUST)
THROUGH:	**DVC (Academic, Student Affairs & Research)**, JOOUST
FROM:	**Dr. Onjala**, JOOUST
DATE:	October 26, 2023
SUBJECT:	**REQUEST FOR APPROVAL OF FUNDING FOR THE DEVELOPMENT OF LUOLINGUAAI PLATFORM (PHASE 1)**

1.0 BACKGROUND

The **LuoLinguaAI** platform is a digital ecosystem designed to preserve, document, and promote the Dholuo language and Luo cultural heritage through Artificial Intelligence. This initiative aligns with JOOUST's strategic objectives in Technology-Enabled Learning (TEL) and digital content creation, and builds on existing research and cultural documentation already undertaken at the University.

This Phase 1 development is positioned within the broader context of the NRF-funded AI-enabled digital repository for Luo Indigenous Knowledge Systems, which covers Homa Bay, Kisumu, and Siaya counties. Phase 1 will therefore be designed with future integration to the NRF repository architecture in mind, ensuring alignment on data models, file naming conventions, and API readiness.

2.0 SCOPE (PHASE 1)

This initial phase will deliver a functional web-based minimum viable product (MVP) including:

- Comprehensive dictionary module (Luo ↔ English)
- Proverbs and riddles repository
- Oral history archive
- Basic administrative dashboard
- Live hosted web portal

Alignment with NRF Repository:

ASPECT	PHASE 1 APPROACH	NRF REPOSITORY STANDARD
Data Models	Will adopt core object structure (Cultural Record, Media Asset, Transcript)	As specified in NRF Technical Specifications
File Naming	Will use deterministic URIs with asset_id and checksums	/JOOUST/NRF/LuoAI_Repository/...

ASPECT	PHASE 1 APPROACH	NRF REPOSITORY STANDARD
API Design	RESTful endpoints with JSON responses	Will align for future interoperability
Metadata	Domain, genre, county/site, consent, restriction	Matches NRF metadata schema

Phase 2 (Future Phase): Mobile application development (Android and iOS) with offline mode, AI Chatbot integration, full gamification module, and deeper integration with the NRF repository. This phase will be proposed upon successful delivery of Phase 1 and will require additional funding.

3.0 DEVELOPMENT TEAM AND BUDGET JUSTIFICATION

The budget of **KES 300,000** covers a lean development team of two (2) personnel over a six-week period, plus essential infrastructure costs. The breakdown is as follows:

ROLE	QUANTITY	DURATION	DAILY RATE (KES)	TOTAL (KES)
Full Stack Developer	1	30 days	5,000	150,000
UI/UX Designer + QA	1	20 days	4,000	80,000
Subtotal - Personnel				**230,000**
Hosting & Infrastructure				40,000
Contingency				30,000
GRAND TOTAL				**300,000**

Justification:

- **Full Stack Developer (KES 150,000):** Responsible for backend development, database architecture, API integration, frontend web development, and deployment. This developer will also ensure alignment with NRF data models and naming conventions. This is a senior-level developer required to deliver a functional platform within the tight six-week timeline.
- **UI/UX Designer + QA (KES 80,000):** Responsible for user interface design across all modules, user experience testing, and quality assurance to ensure the platform is functional and user-friendly.
- **Hosting & Infrastructure (KES 40,000):** Covers cloud hosting services, domain registration, database hosting, and SSL certificate for secure deployment.
- **Contingency (KES 30,000):** For unforeseen expenses, additional server resources, or minor scope adjustments during development.

4.0 TIMELINE

The development will be completed within six (6) weeks from the date of funding approval, leveraging the existing Dholuo dataset already compiled through prior research efforts at JOOUST.

WEEK	ACTIVITY
Week 1–2	Database setup, backend architecture, user management, NRF alignment planning

WEEK	ACTIVITY
Week 3–4	Dictionary module, proverbs and riddles repository
Week 5	Oral history archive, admin dashboard
Week 6	Integration, testing, deployment, and training

5.0 STRATEGIC VALUE

This project will:

- Position JOOUST as a leader in indigenous language preservation and AI innovation
- Generate valuable intellectual property and research assets
- Support the University's digital learning agenda
- Create a scalable foundation for additional Kenyan languages
- Provide a platform for future research grants and international collaboration
- Serve as a public-facing access point for the NRF-funded repository

6.0 REQUEST FOR APPROVAL

Approval is sought for the allocation of **KES 300,000 (Three Hundred Thousand Shillings)** to facilitate the immediate development of Phase 1 of the LuoLinguaAI platform within the proposed six-week timeline.

> **Important Note:** Phase 1 cannot be realised without funding. The costs cover essential personnel, hosting, and infrastructure that cannot be provided in-kind by the University. Successful delivery of Phase 1 will position the University to secure additional funding for Phase 2 and deeper integration with the NRF repository.

7.0 APPROVAL AND SIGN-OFF

ROLE	NAME	SIGNATURE	DATE
Prepared By	Dr. Onjala		
Recommended By	DVC (Academic, Student Affairs & Research)		
Approved By	Vice-Chancellor		

Attachment: Product Requirements Document (PRD)